



Response duration tracks confidence

Hanbei Zhou^{a,b,1}, Rui Zhe Goh^{a,b,1}, Ian Phillips^{a,b,1,2} , and Chaz Firestone^{a,b,1,2} 

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The timing of behavioral responses is among the most robust signatures of cognitive processing—with easier, more accurate, and more confident judgments typically made more quickly. But which aspects of response timing carry these signals? Whereas the time taken to *initiate* a response (e.g., the interval between stimulus onset and keypress) is one of the most popular and informative measures in experimental psychology, another aspect of response timing is its *duration* (e.g., how long the key is held down before being released). Here, across three preregistered experiments spanning perceptual and higher-cognitive tasks, we demonstrate that response duration reliably predicts subjective confidence, over and above response initiation time. When detecting faces (Experiment 1), judging motion coherence (Experiment 2), and answering trivia questions (Experiment 3), subjects held down a response key longer during trials in which they subsequently reported lower confidence in their judgments, as if making these judgments in a tentative fashion. In all cases, response duration accounted for variance in confidence that was not predicted by response initiation time. Our results suggest that response duration itself carries independent information about cognitive processing, advancing a useful and ubiquitously available tool to complement traditional response time measures in the study of cognition.

metacognition | visual perception | high-level cognition | timing | confidence

A foundational principle of experimental psychology is that the timing of external behaviors contains information about internal mental processes. This insight can be traced to at least 1868, when Franciscus Donders used the timing of behavioral responses to isolate different component processes of decision-making (1, 2). Ever since, response timing has served as one of the most prominent measures in psychology, used to investigate phenomena ranging from basic motor skills (3) through attention allocation (4) to higher-level cognition (5).

Yet, response timing is more complex than it may appear. A typical response, such as a keypress, has at least two components: response initiation (pressing down the key) and response completion (releasing the key; Fig. 1). Most research using response timing focuses on *response initiation time*, i.e., the interval between stimulus appearance and response initiation. In contrast, the interval between response initiation and response completion—*response duration (RD)*—has been almost entirely neglected as a measure of cognitive processing. Such neglect is especially significant given the ubiquitous availability of this measure (response duration can be collected in any experiment with keypresses, without additional time or specialized equipment; cf. 6, 7), as well as a recent demonstration that response duration varies independently of response initiation time (8).

These observations raise a crucial question: Does response duration carry meaningful information about the mind? Everyday experience suggests it might. For example, underconfidence in one's judgments may manifest not only in taking longer to initiate a response but also in taking longer to complete it; consider hesitating before releasing a chess piece, or delaying lifting one's finger when clicking "Send" on a high-stakes message. Does response duration provide reliable and independent information about confidence in our responses? (Fig. 2)

The Present Studies: Response Duration and Confidence. Here, across three experiments ranging from lower-level sensory processing to higher-level cognition, we ask whether response duration predicts confidence over and above response initiation time. Subjects made a simple judgment or decision (e.g., detecting motion, answering a trivia question), and indicated their response with a keypress. Following this, they rated their confidence in their answer. All three experiments revealed that response duration (i.e., how long subjects held down the key) reliably predicted confidence, independent of response initiation time.

Results

Experiment 1 comprised a face detection task. On each trial, a field of white noise either contained or did not contain a face, and subjects judged whether a face was present using a keypress. We recorded the duration between stimulus presentation and the first KeyDown

Author affiliations: ^aDepartment of Psychological and Brain Sciences, Johns Hopkins University, Baltimore, MD 21218; and ^bDepartment of Philosophy, Johns Hopkins University, Baltimore, MD 21218

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¹To whom correspondence may be addressed. Email: hanbeizhou@g.harvard.edu, rgoh1@jhu.edu, ianphillips@jhu.edu, or chaz@jhu.edu.

²I.P. and C.F. contributed equally to this work.

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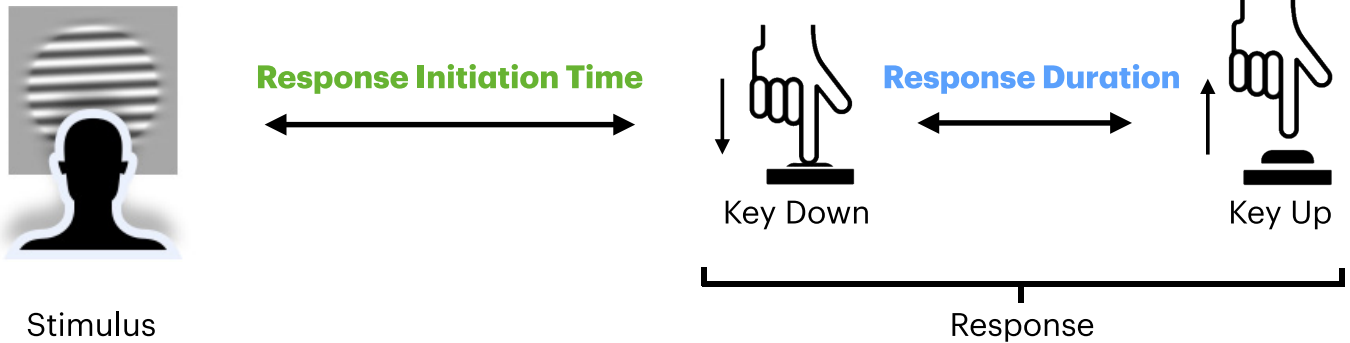


Fig. 1. Response Initiation Time and Response Duration. A typical response, such as a keypress, consists of two stages: response initiation (i.e., KeyDown) and response completion (i.e., KeyUp). Response Initiation Time—often simply referred to as “response time” in the literature—is the interval between stimulus appearance and response initiation. In contrast, Response Duration (RD) is the interval between response initiation and response completion.

event as response initiation time, and the duration between KeyDown and KeyUp events as response duration. After responding, subjects rated their confidence on a 1 to 5 scale, where 1 indicated low confidence and 5 indicated high confidence. Demonstrations are available at <https://perceptionresearch.org/responseduration>.

As per our prediction, a linear mixed-effects model with random intercepts for subjects revealed a significant negative relationship between response duration and confidence ($\beta = -0.000908$, $SE = 0.000157$, $t = -5.79$, $P < 0.0001$): Subjects held the key down longer when less confident in their response. This relationship remained significant after accounting for response initiation time by adding it as a regressor; although response initiation time also predicted confidence as expected ($\beta = -0.000033$, $SE = 0.00000130$, $t = -25.59$, $P < 0.0001$), response duration captured variance in confidence that was not explained by this traditional response timing measure ($\beta = -0.000928$, $SE = 0.000155$, $t = -5.99$, $P < 0.0001$). Furthermore, we replicated the independence (i.e., lack of relationship) between response initiation time and response duration found by Pfister et al.: mean Pearson $r = 0.0128$, $SE = 0.0154$, $t(193) = 0.803$, $P = 0.423$. Finally, a model including both response duration and response initiation time predicted confidence more accurately than a model with response initiation time alone, $\chi^2(1) = 35.90$, $P < 0.0001$. These results demonstrate that response duration tracks confidence independently of response initiation time.

Experiments 2 and 3 generalized our findings. Experiment 2 tested whether the link between response duration and confidence holds for other perceptual tasks. An independent group of subjects completed a motion detection task: Subjects saw an array of moving dots and had to decide whether, as a group, the dots were moving upward. As in Experiment 1, subjects gave their answer with a keypress, and then reported their confidence. Again, response duration correlated negatively with confidence ($\beta = -0.000835$, $SE = 0.000227$, $t = -3.67$, $P < 0.0005$). Moreover, this relationship remained significant when controlling for response initiation time ($\beta = -0.000899$, $SE = 0.000220$, $t = -4.08$, $P < 0.0001$). Thus, the relationship between response duration and confidence is not task-specific but generalizes more broadly.

Experiment 3 generalized our findings beyond perception, to a more cognitive task. On each trial, subjects were given the names of two American cities and estimated which had a larger population. Once again, response duration negatively predicted confidence: Subjects held down the response key longer when less confident in their choice ($\beta = -0.001279$, $SE = 0.000265$, $t = -4.82$, $P < 0.0001$). Furthermore, this effect held when controlling for response initiation time ($\beta = -0.001419$, $SE = 0.000255$, $t = -5.56$, $P < 0.0001$).

As in Experiment 1, Experiments 2 and 3 each found that i) response duration and response initiation time were independent,

and ii) a model including both response duration and response initiation time predicted confidence more accurately than a model with response initiation time alone (*SI Appendix*).

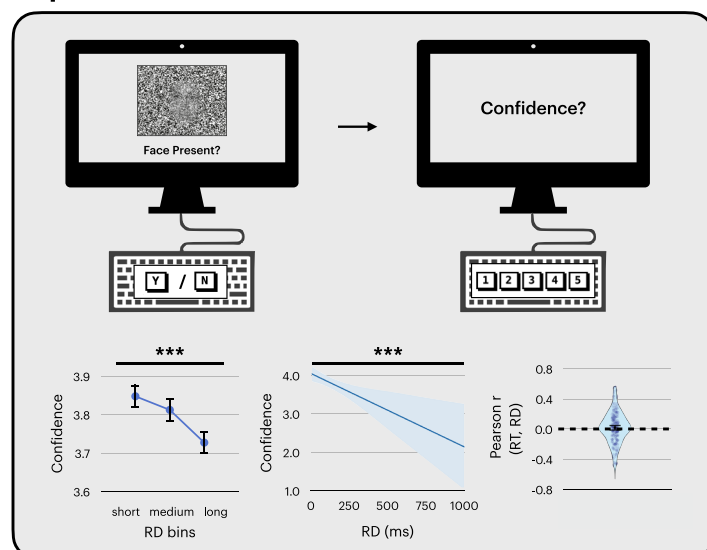
Discussion

The present work demonstrates that a previously neglected aspect of response timing—the duration of responses—contains meaningful information about mental processing. Across three experiments spanning perceptual and cognitive tasks, the duration subjects held down a response key (rather than the time taken to initiate a response) reliably and independently predicted subjective confidence in the response itself.

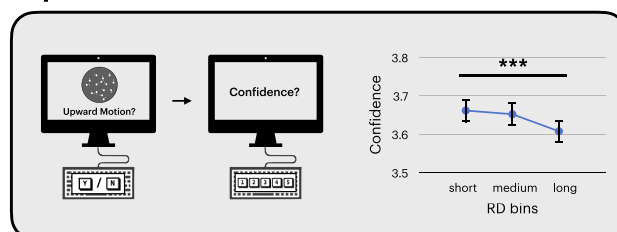
Our findings illustrate how a familiar and ubiquitous aspect of psychological research—pressing a key in response to a stimulus—is not a stereotyped or inflexible event. Rather, this behavior carries information about higher-level psychological factors, in this case subjective confidence (cf. 9). In this way, our findings complement related work linking confidence to movement kinematics, such as the speed of one’s finger when picking up a marble or pressing a touchscreen (6, 7).

Our results also highlight the potential *usefulness* of response duration as a measure in psychology, perhaps especially for studying metacognition (10). Response duration is ubiquitous and freely available: It can be collected in any experiment with keypresses, without any additional burden of time, instruction, or specialized equipment—“a flexible and no-cost tool for psychological science” (8). (Indeed, some past datasets will already include measurements of response duration, providing a potentially rich and untapped source of extant data.) In certain contexts, response duration may even have advantages over traditional measures. Unlike retrospective confidence ratings solicited after a judgment, response duration tracks confidence at (or extremely close to) the moment of judgment, making it less cognitively demanding and perhaps less susceptible to postdecisional factors. Response duration can also easily be collected in populations for whom verbal or ratings-based confidence measures are inapplicable, such as nonhuman animals or prelinguistic infants. Furthermore, in tasks where response initiation time is confounded with other experimental factors (or indeed is itself the key measure), response duration can serve as an implicit measure of confidence. Consider, for instance, visual search for a target among distractors; in such tasks, whereas response initiation time mostly reflects the duration of the search itself—i.e., the time taken to find the target—response duration remains independent of search time and so may offer a unique signal of postsearch cognitive processes. Similarly, in temporal cognition research, timing tasks such as interval-production-by-waiting require subjects to

Experiment 1



Experiment 2



Experiment 3

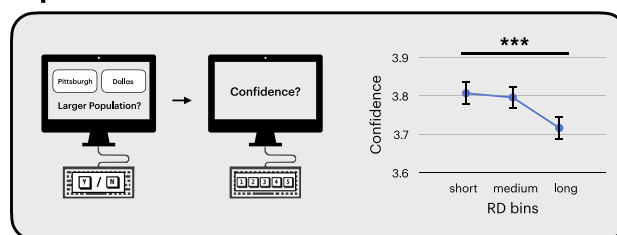


Fig. 2. Stimuli, Procedures, and Results. *Left:* In Experiment 1, a field of white noise either contained or did not contain a face, and subjects judged whether the face was present with a keypress. For visualization purposes, we grouped each subject's response durations into short, medium, and long bins; the average confidence for each bin decreases as response duration increases. A mixed-effects model with subjects as random intercepts revealed a correlation between confidence and response duration, with confidence decreasing as response duration increases. There is no significant correlation between response initiation time (RIT) and response duration. *Top right:* In Experiment 2, subjects saw an array of moving dots, and had to decide whether, as a group, the dots were moving upward. Again, confidence decreases with the increase in response duration. *Bottom right:* In Experiment 3, subjects saw the names of two American cities and estimated which had a larger population. Response duration correlated negatively with confidence. All error bars represent 95% confidence intervals.

initiate a response after a specified period has elapsed (e.g., 11); in such cases, response initiation time primarily reflects a temporal judgment, whereas response duration offers a potentially unique signal of nontemporal cognitive processing.

An important question for future research is the directionality of the present effects. It is natural to suppose that response time *reflects* confidence, such that we hesitate more because we are less confident. But confidence may also be influenced by response duration, such that we are less confident because we hesitate more (12, 13). Importantly, either possibility permits response duration to prove useful as described above.

While the present work focuses on confidence, one could also explore whether response duration carries information about other aspects of the mind. For example, response duration might reflect

the speed of evidence accumulation. Indeed, one potential explanation of the relationship between response duration and confidence is that evidence accumulation continues after action initiation until a second action termination threshold is reached (cf. 14). Finally, an exciting avenue for future work would be to explore response duration in subjects other than human adults.

Methods

Supplementary methods appear in *SI Appendix*. Studies were approved by the Johns Hopkins University Institutional Review Board. All subjects gave informed consent.

Data, Materials, and Software Availability. Anonymized Data, code, materials data have been deposited in Open Science Framework (<https://osf.io/srtf4>) (15). All other data are included in the manuscript and/or *SI Appendix*.

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